MTH 201 Chapter 1: Application of Integration

1.4 Arc Length Worksheet

1. Find the exact arc length of the curve over the given interval: $y = x^2$ from $x = 1$ to $x = 3$.

(a) Set up, *but do not evaluate*, an integral with respect to *x*.

(b) Set up, *but do not evaluate*, an integral with respect to *y*.

2. Determine the length of $y = \ln(\sec(x))$ between $0 \le x \le \frac{\pi}{4}$.

(Hint: Although we have not shown this in class yet it is true that $\int \sec(x)\,dx = \ln|\sec(x) + \tan(x)| + C$)

3. Setup, but do not evaluate, an integral which represents the length of the curve $y = \sqrt{x}$ on $[0,4]$.

4. Find the exact arc length of the curve over the given interval. $x = \frac{1}{3}\left(y^2 + 2\right)^{3/2}$ from $y = 0$ to $y = 1$.

5. Find the exact arc length of the curve over the given interval. $y=\dfrac{x^6+8}{16x^2}$ from $x=2$ to $x=3$.

6. Kyle is setting up an integral with respect to y which is supposed to find the arc length of the curve $y=x^3+4$ on $[-1,1]$. He sets up the following integral with respect to y

$$L=\int_3^5 \sqrt{1+\left(\frac{1}{3}(y-4)^{-2/3}\right)^2}\,dy$$

7. Find the exact arc length of $24xy = y^4 + 48$ from $y = 2$ to $y = 4$.

8. Find the exact arc length of $y = 3x^{3/2} - 1$ from $x = 0$ to $x = 1$.